

FILTER DESIGN TEST CASE

High-Pass Filter - Band 20 (800 MHz) for GPS/GNSS Anti-Jamming Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a High-Pass Filter filter targeting Band 20 (800 MHz) for GPS/GNSS Anti-Jamming Filter. The filter is designed on PZT on Si substrate with a target center frequency of 3317.1 MHz and bandwidth of 57.2 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the GPS/GNSS Anti-Jamming Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: High-Pass Filter
- Frequency Band: Band 20 (800 MHz)
- Application: GPS/GNSS Anti-Jamming Filter
- Substrate: PZT on Si
- Package: SiP (System in Package)
- Design Tool: PathWave EM Design
- Design Center: Newbury Park Design Center

This specification applies to all variants of the High-Pass Filter design targeting Band 20 (800 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	3317.1 MHz
Bandwidth (3 dB):	57.2 MHz
Insertion Loss Target:	≤ 1.2 dB
Return Loss Target:	≥ 21.0 dB
Out-of-Band Rejection:	≥ 36.0 dB
Isolation (Duplexer):	≥ 28.0 dB
Q Factor:	10822
Temperature Coefficient:	-17.9 ppm/°C
Die Size:	0.8 x 1.4 mm
Wafer Size:	6-inch
Number of Resonators:	11
Electrode Material:	Mo

Passivation: Si3N4

4. TEST PROCEDURE

1. Open PathWave EM Design and load the High-Pass Filter template project.
2. Define the substrate stack: PZT on Si with specified layer thicknesses.
3. Set design targets: center freq 3317.1 MHz, BW 57.2 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.2 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (SiP (System in Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, PZT on Si).
10. Perform on-wafer probing using Anritsu MS46122B ShockLine.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 9951 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.2 dB
- Return loss in passband shall be >= 21.0 dB
- Out-of-band rejection at +/- 86 MHz offset >= 36.0 dB
- Group delay variation across passband shall be <= 7 ns
- Temperature drift over -40°C to +85°C shall be <= 7.4 MHz
- Power handling shall be >= +33 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 87%

6. TEST RESULTS

Measurement Date: 2024-03-21
Devices Tested: 75
Insertion Loss (meas): 0.86 dB
Return Loss (meas): 22.6 dB
Rejection (meas): 38.6 dB
Isolation (meas): 32.8 dB
Q Factor (meas): 10822
Group Delay Variation: 7.9 ns
Power Handling: +30 dBm
Die Yield: 85.3%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

- a) The High-Pass Filter design demonstrated excellent performance for Band 20 (800 MHz).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Temperature stability was within specification across full range.
- e) Wafer-level uniformity was excellent (sigma < 1.5%).

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

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